

Maximum Trading Opportunity

Calculation of the maximum trading opportunity is as follows:

Let r be the current portfolio and z_k indicate the number of shares to trade in stock k .

Then these vectors are:

$$r = \begin{pmatrix} r_1 \\ \vdots \\ r_k \\ \vdots \\ r_m \end{pmatrix} \quad z = \begin{pmatrix} 0 \\ \vdots \\ z_k \\ \vdots \\ 0 \end{pmatrix} \quad (r - z) = \begin{pmatrix} r_1 \\ \vdots \\ r_k - z_k \\ \vdots \\ r_m \end{pmatrix}$$

where $(r - z)$ is the portfolio after trading z_k shares of stock k . Notice that the z vector only contains one value z_k for the stock that we are looking to trade. Having zeros in the other entries ensures that the other position sizes remain constant.

Our goal here is to determine the maximum number of shares z_k that can be traded such that the risk after trading is equal to the risk before trading.

Mathematically, this is as follows:

$$\sqrt{r'Cr} = \sqrt{(r - z)'C(r - z)}$$

Squaring both sides yields:

$$r'Cr = (r - z)'C(r - z)$$

Expanding this equation yields:

$$z'Cz - 2r'Cz = 0$$

Solving for z_k yields two solutions as is expected since this is a quadratic equation. These solutions are:

$$z_k = 0 \tag{9.28}$$

$$z_k = 2 \cdot \frac{1}{\sigma_k^2} \sum_{j=1}^n r_j \sigma_{ij} \tag{9.29}$$

The first solution (Equation 9.28) is the naïve solution and implies that we do not trade. If there are no transactions then of course the risk does not change. The second solution (Equation 9.29) is the value that is most interesting to traders. It signifies the most trades that can occur without adversely affecting portfolio risk. Also notice that this solution is twice